

(MODEL QUESTION)

TED(15) 4026

Register No -----

REVISION 2015

Signature -----

FOURTH SEMESTER DIPLOMA IN MECHANICAL ENGINEERING

PRODUCTION DRAWING

(Maximum marks : 100)

PART A

(Maximum marks : 20)

I Answer **ALL** questions. Each questions carries 5 marks

1. Define and sketch different types of fits used in assembly drawing.
2. Draw the surface texture symbol and mark the following characteristics on it.
 - a. Roughness value
 - b. production method
 - c. Sampling length
 - d. Direction of lay
 - e. Machining allowance
3. State different types of process charts used in a manufacturing industry.
4. Express the numerical values of the following fits .
 - a. H7js6 b. H7e8

PART B

(Maximum marks : 30)

II Answer any **TWO** questions from the following. Each question carries 15 marks

1. Prepare an operation process chart of a **locating pin** shown in **fig.1** incorporating the following details

Length $L = 50$, Diameter $D = 10$ $L_4 = 3$, $D_3 = 2$

Part name : Locating pin

Part No : 2400 14 61

Drawing No : LP 00023 45

Equipment : Drill jig

Material : steel

Specification : IS 656 part 2

Qty required : 50 Nos

Also mention the department, machines ,tools/gauges, Weight per piece, Total number of operation cycle time etc

2. Compute limit dimensions for a clearance fit on hole basis system

Basic size $\phi 50\text{mm}$

Minimum clearance 0.05mm

Tolerance on hole 0.035 mm

Tolerance on shaft 0.02mm

Check the calculated dimensions and represent schematically

3. Elevation of a **Go and No go** gauge is shown in **fig.2**.The surfaces are identified by the numbers 1 to 7 and the surface roughness values of these surfaces are given below. Redraw the given figure and indicate surface roughness values using grade numbers as per BIS

surfaces	1	2	3	4	5	6	7
Roughness		0.4	12.5	6.3	12.5	0.4	1.6
Values in μm	1.6						

PART C

III. Answer any **ONE** Question from the following. Each question carries 50 marks

1. An assembly drawing of a **connecting rod** is given in **fig.3** Prepare a shop floor drawing for the production, incorporating the following details
 - a. Small end of the connecting rod is connected with the Gudgeon pin with an easy running fit
 - b. Big end of the connecting rod is connected with crank shaft with a loose running fit
 - c. Bushes inside the big end and small end are connected with a push fit
 - d. The axis of the big end and small ends are parallel within the limits of 0.04mm
 - e. All the mating surfaces are to be finished with a roughness value of 3.2 μ
 - f. All the other surfaces are to be finished with a roughness value of 6.3 μ
2. Prepare a shop floor drawing for the production of **Socket and Spigot Joint** is shown in **fig: 4** , incorporating the following information with item list.
 - (a) Socket and Spigot are to be manufactured with an easy running fit.
 - (b) Cotter pin is to be fixed with a sliding fit.
 - (c) A parallelism tolerance is to be given to the inside end of spigot and socket with the limit of 0.04mm.
 - (d) The bore of the socket and outside surface of the spigot are to be manufactured co-axial to a tolerance value of 0.03mm.
 - (e) Inside end of the spigot is perpendicular to the axis with a tolerance of 0.05mm
 - (f) All the mating surfaces are to be finished to a roughness value of 3.2 microns.
 - (g) All the other surfaces are to be finished with a roughness value of 6.2 microns.

(1X50=50)

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Form A

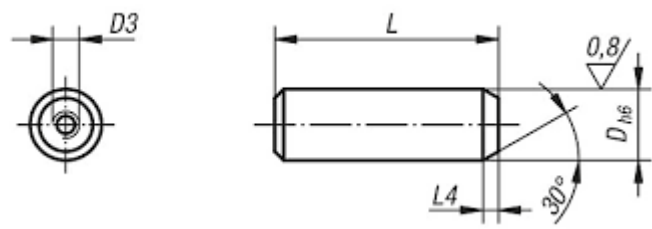


FIG-1

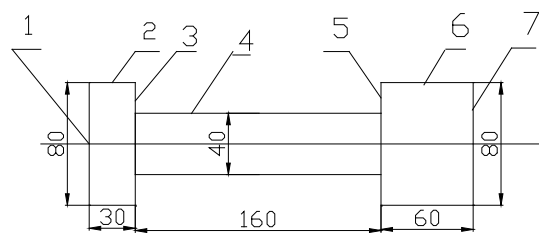
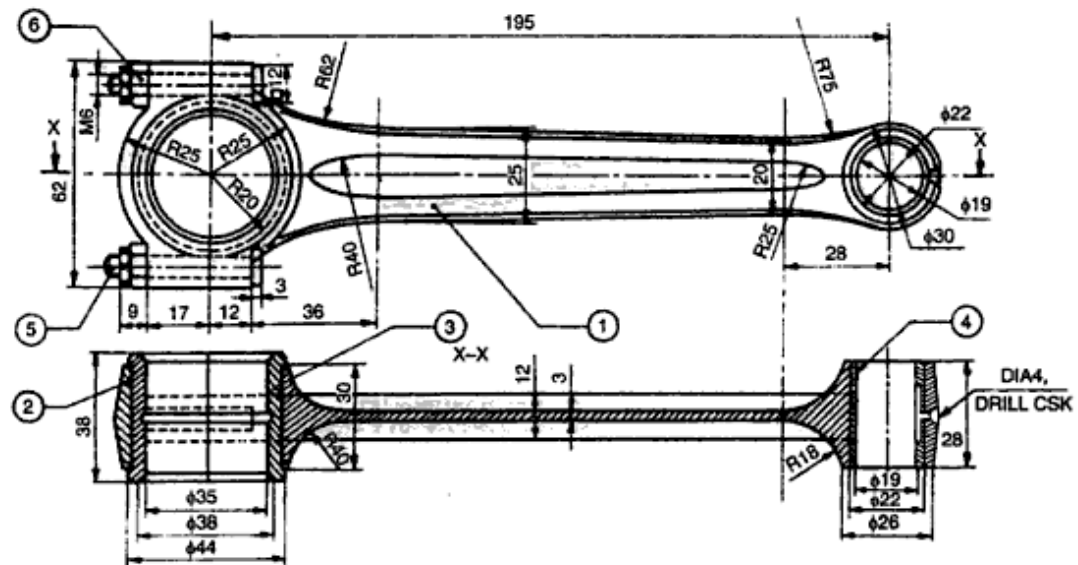


FIG 2



Parts list

Part No.	Name	Matl.	Qty.
1	Rod	FS	1
2	Cap	FS	1
3	Bearing brass	GM	2
4	Bearing bush	P Bronze	1
5	Bolt	MCS	2
6	Nut	MCS	2

FIG 3

Cotter Joint with Socket and Spigot ends

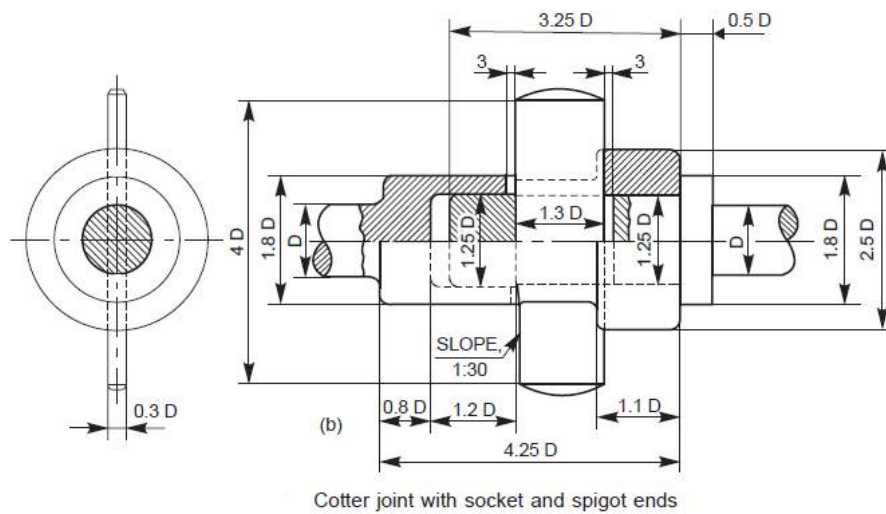


FIG 4